



Birla Institute of Technology & Science, Pilani

Pilani Campus

ACADEMIC— UNDER GRADUATE STUDIES DIVISION

Instruction Division

First Semester 2026-27 Course Handout Part II

Date: 04-08-2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No.: ME F317
Course Title: Engines, Motors, and Mobility
Instructor In-charge: Prof. A R Harikrishnan
Tutorial Instructors: Prof. Shyam Sunder Yadav, Prof. Chennu Ranganayakulu, Prof. Suvanjan Bhattacharyya, Mr. Krishnaram K K

Course Description:

The internal combustion engine (IC) is a heat engine that converts chemical energy in a fuel into mechanical energy, usually made available on a rotating output shaft. This is one of the dominating areas in prime movers. These engines are often used for the propulsion of a vehicle (i.e., automobile, truck, locomotive, marine vessel, or aeroplane), and other applications include stationary engines to drive generators or pumps and portable engines for things like chain saws and lawnmowers, etc. This course will introduce the students to system-level design concepts and the workings of internal combustion engines. Furthermore, electric mobility has become an essential part of the energy transition. This course focuses on the fundamentals of the technology behind electric vehicles. It will cover the contents such as the working principle of electric vehicles and delve into its key components, e.g. motors and power electronics, battery technology, charging technologies and future trends in the development of electric vehicles.

Scope and Objective of the Course: This course has been designed to make the students familiar with internal combustion engines and electric and hybrid electric power trains.

Text Books:

V. Ganesan, **Internal Combustion Engines**, Tata McGraw-Hill Pub. Co. Ltd, New Delhi, 4th Edition.

Reference Books:

1. M.L. Mathur, R.P. Sharma, “**A Course in Internal Combustion Engines**”, Dhanpat Rai and Sons, New Delhi, 2005.
2. H.N. Gupta, “**Fundamentals of Internal Combustion Engines**”, Prentice Hall of India, 2nd Edition, 2013.



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3. John B. Heywood, “**Internal Combustion Engine Fundamentals**”, McGraw-Hill Education Private Limited, 2011
4. Mehrdad Ehsani, Yimin Gao, Stefano Longo, Kambiz M. Ebrahimi, “**Modern Electric, Hybrid Electric, and Fuel Cell Vehicles**”, CRC Press Taylor & Francis Group, 3rd Edition, 2018.

Course Plan:

Module	Lecture	Learning Objective	Topic to be discussed/ Learning outcomes	Ref. to text
Introduction to IC Engines, motors and mobility	1	Concept of IC engines, classifications, working, concept of electric vehicles and mobility	What is Engine, What is IC engine? Different Types of Engine, Electric and hybrid vehicles Future mobility	TB Ch-1, class notes
	2-3	Components of IC Engines and geometrical parameters, working principle of the engines	Understanding the IC engine components and their working.	TB Ch-1, class notes
	4	Engine Performance Parameters	- Geometrical parameters of the engine. - Understand the significance of each performance parameter The effect of these parameters on the Engine performance	TB Ch-1, class notes
Air standard Cycle, Fuel-Air Cycles and their Analysis	5	Air standard Cycle: Otto Cycle Diesel Cycle and Dual Cycle Fuel-air cycle, factors affecting the cycle	- Understanding the concept of ideal cycles, performance of a real SI and CI engine through the idealized closed cycle. - Understand effect of fuel-air mixture on the performance, study the factors affecting the performance.	TB Ch-2 and 3
	6	Comparison and analysis of Air standard cycle with fuel–air cycle.	- Comprehend the difference between the Air standard cycle and the Fuel-air Cycle. - Understand the effect of dissociation, specific heat on the higher temperature and pressure of the Otto and Diesel cycle.	TB Ch-3
Actual cycles and analysis	7	Deviation of actual cycle from the ideal cycle	Comprehend the difference between actual and air-standard cycle.	TB Ch-4
	8	Factors influencing the performance	Understand the effect of different factors like time loss, heat loss, exhaust blow down etc. on the ideal cycle.	TB Ch-4



Conventional and Alternate Fuels	9	Different conventional fuels like paraffin, olefin etc, Important qualities of the fuel and rating of the fuel	Study the different types of fuel which are in use and their properties, understand the important qualities of the SI and CI fuel as well as they will understand the rating of the fuel	TB Ch-5
	10-11	Different alternative fuels and their properties	- Need of alternative fuels and their suitability - Different possible alternative fuel like natural gas, hydrogen, alcohols etc.	TB Ch-6
Carburetion and Fuel Injection in S.I. Engines	12-13	Mixture formation requirements for SI engines, Carburettor working principle, Fuel injector	- What is a Carburettor, and why is it required, Mixing of fuel and air in the carburetion - Study the mechanism for controlling the air-fuel ratio. - How a gasoline fuel injector works	TB Ch7, RB2 Ch-9
Fuel Injection in C.I. Engines	14	Injection requirements for CI engine, Injection systems, spray characteristics Pumps, Injectors, Nozzles, Electronic control	Understand the importance of fuel injection as it affect the engine performance and the power output. Understand the different element in the fuel injection system	TB Ch-8,9, RB2 Ch-9
Combustion in S.I. and C.I. Engines	15	Normal and abnormal combustion, factors affecting ignition lag and combustion process	Homogeneous and heterogeneous mixture, Understand the combustion process in an SI engine, Understand the ignition delay in the SI engine combustion process	TB Ch11, RB2 Ch-6-7, Class note
	16	Combustion chambers for S.I. engines, Knocking and Detonation	- Understand the reason for abnormal combustion and the phenomenon of knocking and detonations. - Appropriate design of the combustion chamber for SI engine	
	17	Combustion in CI engines, Heat release rate, Air-fuel ratio, Knocking, Factors affecting combustion	- Will comprehend the different stages of the combustion process in the CI engines. - Factor affecting the combustion process, such as ignition delay. - How abnormal combustion occur in a CI engine, and how to avoid it.	
	18	Combustion chamber design for CI engines, Direct and Indirect injection, Air flow motion, Swirl	Will understand the need of combustion chamber design for proper mixing of the fuel in the air. Study the different flow pattern of the air in order to mix fuel with air.	



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Emissions	19	Heat transfer and Cooling requirements of different components, Different cooling systems	- Understand the importance of cooling for safe operation of the engine e.g. in order to avoid the chemical and physical changes in the lubricating oil which may cause wear and sticking of the piston rings. - Study the different cooling system and mechanisms.	TB Ch13, RT2 Ch-14, class notes
	20-21	Exhaust system, emissions from IC engines, emission norms, emission control strategies.	What are the important pollutants from the IC engines? What are the emission norms and their importance? What are the emission control techniques used in the IC engine vehicles?	
Performance testing of IC Engines	22	Measurement of brake power, Indicated power, Sp. Fuel consumption, performance characteristics of engines	- Study the different performance parameters of the IC engine. - Study the energy management of the IC engine.	TB Ch16, RT2 Ch-16
Introduction to electric and hybrid electric vehicle power trains	23-24	Electric and hybrid electric vehicle architectures, functioning of each components, performance analysis and comparison	- IC engine Vs electric power train - Hybridization of IC engine power train - Performance comparison of IC engine and electric power train	Class notes, RB 4 Ch- 5-6
Battery Technology for electric vehicle	25-26	Different storage technologies for electric vehicle, Li-ion batteries, Battery working principles, Battery losses, Battery pack and battery management system	What are the different energy storage options available for electric vehicles? How a Li-ion battery works? Performance assessment of a battery. How to design a battery pack for electric vehicle? How a battery management system works?	Class notes
Motors and Controllers: Fundamentals and Design	27-28	Electric Propulsion Systems: various motor drives, their functioning and basic selection criteria; power electronics, regenerative braking	What are the different types of motor drives used in electric vehicles? How to select a suitable motor for electric vehicle?	Class notes, RB 4 Ch- 7

Evaluation Scheme

Components	Duration	Weightage (%)	Date & Time	Remarks
Mid-semester exam	90 min.	30	07/10 AN1	Closed book
Minor projects/ Case study	4-5 weeks	10	Continuous	Open book



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Quiz	30 min	20	Announced- Refer to the tentative dates mentioned below	Open book
Comprehensive exam	3 hrs.	40	5/12 FN	Partially closed and partially open book

Notices: All notices related to this course will be updated on the Nalanda. The students should also check their emails regularly.

Make-up Policy: It will be as per the institute rules. Only genuine cases will be considered. No makeup will be provided for the quizzes under any circumstances.

Quizzes: There will be quizzes based on the objective/subjective type questions during the tutorial hours. The best four out of five quizzes will be taken for grading. Tentative quiz dates are: 25th August 2025, 8th September 2025, 22nd September 2025, 27th October 2025, and 10th November 2025. If there are any changes to quiz dates, they will be communicated in advance. **No makeup will be provided for these quizzes under any circumstances.**

Minimum Passing Marks: The passing marks for this course will be determined based on one of the two criteria: 30% of the median of the post-comprehensive examination score or 20% of the average of the first three top scorers. The lower value between these two criteria will be considered as a passing mark in this course.

Chamber consultation hour: Monday: 5.30 PM-6.30PM

Instructor-in-charge

Prof. A R Harikrishnan

ME F317



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